

Chapter 6-8 cvd pvd 外延

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1.Lithography determines chip feature size. Feature size determines integrated circuit performance and size.

$$R = \frac{k_1 \lambda}{NA}$$

you can choose to imcrease Na or decrease λ or K_1

2.Immersion technology: Immersion technology can improve resolution and pattern fidelity by injecting a liquid medium (such as water) into the air gap of the lithography system to match the refractive index between the photoresist and the silicon wafer surface, thereby reducing reflection and scattering between the photoresist and the silicon wafer surface.

As for the innovation of integrated circuit manufacturing technology, I think it is one of the key factors to promote the development of the entire electronics industry. With the continuous progress of technology and the constant change of market demand, innovation is the driving forces for the advancement of technology in china in recent years. By constantly improving existing technologies and exploring new manufacturing methods and materials, higher performance, lower power consumption, smaller size microelectronic devices can be realized, and the entire electronics industry can be driven towards a more advanced and sustainable direction.

3.The resolution of the reticle determines the minimum feature size that can be made, while the quality of the lithography affects the accuracy and repeatability of the features. Stepper and Scanner lithography techniques are used in modern lithography processes to achieve high resolution, high quality lithography and to ensure light stability and consistency throughout the manufacturing process.